Modeling and Validation of a Soccer Ball's Three-Dimensional Flight to the Goal

Problem B

Abstract

Gravity bends a soccer ball's path, drag reduces its speed, and Magnus lift redirects a spinning ball. We formulate a three-dimensional model with these three forces and constant spin. Four models distinguish gravity alone, constant drag, constant drag and lift, and complete speed- and spin-dependent aerodynamic coefficients. Published Teamgeist relations reproduce the selected 2007 wind-tunnel drag and lift curves with coefficient RMSE values of 0.0113 and 0.0504.

Forward calculations are compared with controlled launch experiments, including a measured range of 21.9 m and four rotating-ball trajectory segments. The variable-coefficient model predicts a range of 21.871 m in the nonspinning experiment. Inverse calculations estimate initial motion and nonnegative force corrections from two later positions, selecting corrections close to unity. Five recorded match flights provide subsequent positions for comparison. For the competition data, the variable-coefficient model predicts the goal-plane centre $(y, z) = (2.683, 1.959)$ m at $t = 0.4026$ s. A separate same-initial-state calculation quantifies rotational damping, while surface geometry and wind are discussed through published flow results and force estimates.

Keywords: Soccer ball, aerodynamic drag, Magnus effect, inverse prediction, trajectory comparison

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1 Introduction

1.1 Problem Background

The path of a kicked soccer ball is not generally a straight line. Gravity bends the trajectory downwards, air resistance slows the ball, and its rotation can produce a sideways or vertical aerodynamic force. The combination of these effects determines where the ball reaches the plane of the goal. Mathematical models of soccer-ball flight have therefore been developed using Newton's law together with experimental aerodynamic relationships [1, 4, 8]. High-fidelity simulations have also shown how seams and other surface features can alter the flow and the trajectory [5].

An effective way to examine a physical model is to compare its calculated path with positions recorded during a real kick. The DFL open dataset combines match events and optical tracking data for professional matches [2, 3]. These records make it possible to identify a penalty event, follow the ball through the flight, and compare the model with its position at the actual goal plane.

1.2 Problem Restatement

We model a soccer ball travelling in three dimensions from the vicinity of the penalty mark to the goal. The goal is 7.32 m wide and 2.44 m high; the penalty mark is 11 m from the goal line. We establish a model incorporating gravity, drag and the Magnus effect, solve for its trajectory and the point at which it crosses the goal plane, and compare the calculated positions with a recorded penalty flight. The prediction uses the initial position and the positions at 0.05 and 0.10 s; subsequent positions are reserved for validation. Figure 1 summarizes the force models and inverse problem: early positions determine initial conditions and force corrections, while later positions provide validation targets. Section 2 states the assumptions and notation. Section 3 establishes the forces and equations. Section 4 compares forward and inverse predictions. Section 5 is reserved for sensitivity analysis. Section 6 discusses surface geometry, spin damping and wind, followed by conclusions and model evaluation in Sections 7 and 8.

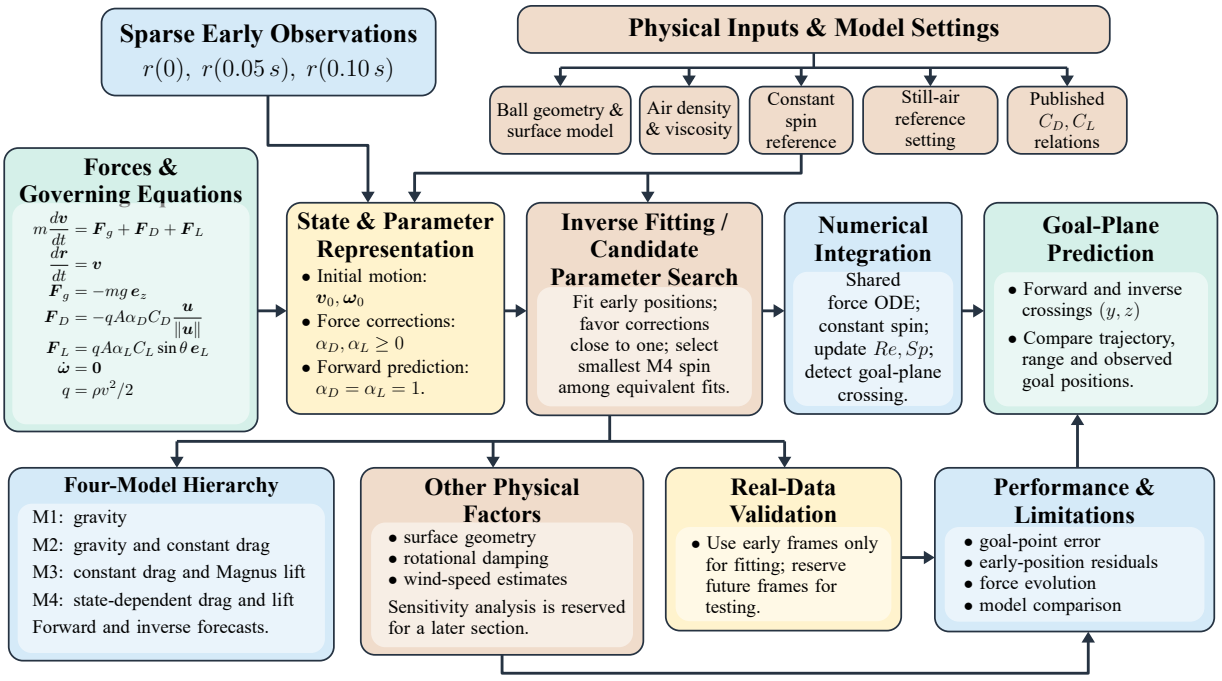


Figure 1: Three-force modeling, four-model comparison and goal-plane validation.

2 Preparation of the Models

2.1 Assumptions

- The ball is idealized as a rigid sphere of constant mass and radius, with cross-sectional area $A = \pi R^2$ and position at its centre, following common soccer-ball flight models [4, 1, 6]. Empirical football coefficients represent the average aerodynamic effect of its surface. Section 6.1 discusses the effect of seams and asymmetric geometry.

- Gravity, aerodynamic drag and spin-induced lift act during free flight [4, 1]. Integration ends at ground contact or the selected goal plane.

- The main model neglects aerodynamic torque, so its spin vector remains constant. Section 6.2 examines this approximation by adding rotational damping to the same initial state.

- Air density and viscosity are uniform, and the main calculation uses still air. Section 6.3 estimates how wind changes relative airflow and force magnitude.

- The initial position and the next two early positions determine the inverse input. Subsequent positions from the DFL tracking records are used for comparison [2, 3]. The recorded vertical coordinate is interpreted as height in metres.

2.2 Notations

Table 1: Principal quantities and competition reference values.

Symbol	Meaning	Value or unit
$\boldsymbol{r}, \boldsymbol{v}$	Ball-centre position and velocity	m; m/s
$\boldsymbol{\omega}, \omega$	Spin vector and its magnitude	rad/s
m, R, A	Mass, radius and frontal area	0.43 kg; 0.11 m; πR^2
g	Gravitational acceleration	9.81 m/s^2
$\boldsymbol{u}, v$	Air-relative velocity and its magnitude	m/s
ρ	Air density	1.2 kg/m^3
$\nu, \mu = \rho\nu$	Kinematic and dynamic viscosity	1.50×10^{-5} m^2/s; Pa s
Re, Sp	Reynolds number and spin parameter	dimensionless
C_D, C_L	Drag and transverse lift coefficients	dimensionless
α_D, α_L	Multiplicative force corrections	dimensionless
$\theta, \boldsymbol{e}_L$	Spin–airflow angle and Magnus direction	rad; unit vector
X_g	Local goal-plane coordinate	11 m
$\boldsymbol{Q}, I_b$	Damping torque and moment of inertia (Section 6)	N m; kg m^2

The competition mass, radius and gravity follow the reference values of Maponi and Piombin [4]; the air density follows Asai et al. [7]. The viscosity is a declared reference air setting, with $\mu = \rho\nu$. Controlled experiments use their reported mass, diameter and air properties.

3 Model Establishment

The same Newtonian equations generate all four models by selecting the included aerodynamic terms. We first describe the geometry, then connect the forces to the surrounding flow and ball rotation.

3.1 Geometry and Goal Plane

For a pitch 105 m long with its origin at the centre, the two goal lines are located at $x = \pm 52.50$ m. The selected kick travels towards the positive-x goal. For the competition geometry the valid goal opening is $-3.66 \le y \le 3.66$ m and $0 \le z \le 2.44$ m. The goal-plane position is the ball centre's first forward intersection with $x = X_g$.

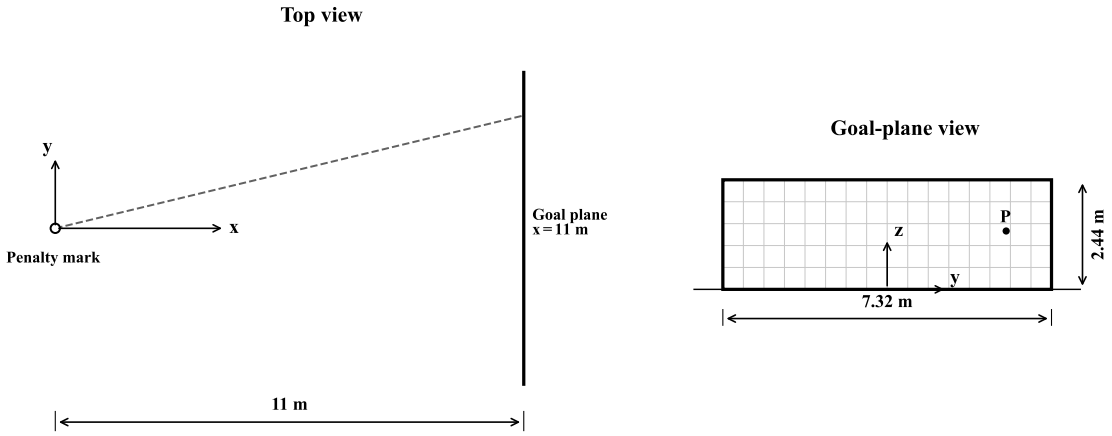


Figure 2: Penalty-kick geometry in local coordinates. The dashed line and point P indicate an illustrative crossing.

Figure 2 uses the penalty mark as the local origin and places the goal plane at $x = 11$ m. For the match record below, the pitch-centred coordinate system instead places the goal at $X_g = 52.50$ m.

The ball crosses the opening cleanly when its centre satisfies $|y| \leq 3.66 - R$ and $R \leq z \leq 2.44 - R$ at $x = X_g$. Ground contact occurs when the descending centre reaches $z = R$. These event definitions are used consistently for all four models.

3.2 Force Analysis

3.2.1 Gravity

Gravity supplies a constant downward force,

$$\boldsymbol{F}_g = -mg\boldsymbol{e}_z, \tag{1}$$

where $\boldsymbol{e}_z$ is the upward vertical unit vector. It produces downward curvature even when all aerodynamic forces are removed.

3.2.2 Aerodynamic Drag

A moving ball transfers momentum to the air in its wake. The mass flow through area A scales as $\rho A v$, and the velocity change scales as v, giving a force scale $\rho A v^2$. Introducing dynamic pressure $q = \rho v^2/2$ gives the conventional drag expression [11, 8],

$$\boldsymbol{F}_D = -\tfrac{1}{2}\rho A \alpha_D C_D(Re, Sp) v \boldsymbol{u}, \qquad \boldsymbol{u} = \boldsymbol{v} - \boldsymbol{v}_{\text{wind}}, \quad v = \|\boldsymbol{u}\|. \tag{2}$$

Its quantities have the following physical roles:

1. **Air density ρ.** The density measures air mass per unit volume. We use the uniform reference value 1.2 kg/m^3 reported by Asai et al. [7].

2. **Projected area $A = \pi R^2$.** A larger radius intercepts more airflow; the spherical geometry fixes this area throughout flight.

3. **Relative speed and direction v, u.** The quadratic speed dependence gives stronger resistance at higher speed, and the negative direction opposes airflow. The still-air main model has $u = v$.

4. **Drag coefficient $C_D(Re, Sp)$.** The coefficient represents boundary-layer transition, separation and wake formation. Following Asai et al. and Kiratidis and Leinweber [7, 8], we define

$$Re = \frac{2Rv}{\nu}, \qquad Sp = \frac{R\omega}{v}, \qquad \omega = \|\boldsymbol{\omega}\|, \quad \nu = \frac{\mu}{\rho}. \tag{3}$$

Re compares inertial and viscous flow effects; Sp compares rotating surface speed with airspeed. Constant spin makes Sp rise as the ball slows.

The drag crisis is the rapid reduction of drag associated with boundary-layer transition. A useful coefficient law should describe this transition as well as the slower variation in the supercritical regime. According to Kiratidis and Leinweber [8], their Eq. (5) represents zero-spin drag by

$$\begin{aligned} f(v) = {} & \frac{a - b_{\text{min}}}{1 + \exp[(v - v_c)/v_s]} + b_{\text{min}} \\ & + \frac{v - v_{\text{min}}}{1 + \exp[-(v - v_{\text{min}})/v_s]} \frac{b_{\text{max}} - b_{\text{min}}}{v_{\text{max}} - v_{\text{min}}}. \end{aligned} \tag{4}$$

The coefficient a sets the low-speed level, v_c locates the transition, and v_s controls its width. The remaining parameters set the level and slope of the high-speed tail. Table 2 gives the Teamgeist parameters reported in their Table 1 [8].

Table 2: Teamgeist zero-spin drag parameters from Kiratidis and Leinweber [8].

a	v_c	v_s	b_{min}	b_{max}	v_{min}	v_{max}
—	m/s	m/s	—	—	m/s	m/s
0.4927	12.58	1.071	0.1440	0.1540	23.17	35.00

For a spinning Teamgeist, Kiratidis and Leinweber's Eq. (7) adds a linear spin correction with slope 0.514, fitted to the supercritical wind-tunnel data [8]:

$$C_D(Re, Sp) = f\left(\frac{Re\nu}{2R}\right) + 0.514Sp. \tag{5}$$

Thus drag changes with speed even at zero spin, and also increases with rotation. The Teamgeist parameter set is used throughout the coefficient comparison; other ball surfaces require their own parameters.

The final M4 trajectory has speeds of 24.99–32.57 m/s and Reynolds numbers of $(3.67–4.78) \times 10^5$. Kiratidis and Leinweber's Teamgeist fit extends to 35 m/s [8], corresponding to $Re = 5.13 \times 10^5$ for the reference air. The competition flight lies in this speed range and in the supercritical regime represented by the empirical relation, supporting its use here. The rotating Asai measurements span $Re = 333793–455172$; the coefficient figures use the

measured upper condition exactly. The inferred competition spin gives $Sp = 0.427$–0.557; its higher values extend beyond the selected wind-tunnel spin interval.

5. **Drag correction $\alpha_D \geq 0$.** This dimensionless multiplier equals one in forward calculations and is estimated from early positions in inverse calculations.

Comparison with published wind-tunnel data. Asai et al. [7] measured soccer-ball forces in Tokai University's low-speed recirculating wind tunnel using a six-component balance (1.5×1 m test section; about 1% turbulence). We use their spinning full-size Teamgeist FRP replica: $R = 0.11$ m, $\rho = 1.2$ kg/m^3, airspeed 22–30 m/s and spin 2–10 rev/s. Five points per coefficient are digitized from their Figures 8 and 9 at $Re = 455172$ (about 30 m/s); the reported side force C_S represents transverse Magnus lift here. The speed–Re correspondence gives $\nu \simeq 1.45 \times 10^{-5}$ m^2/s for this matched comparison.

Figure 3 reproduces the increase of drag with Sp: the coefficient MAE is 0.00985 and RMSE is 0.01127. Both curves use the same experimental condition and measured spin interval.

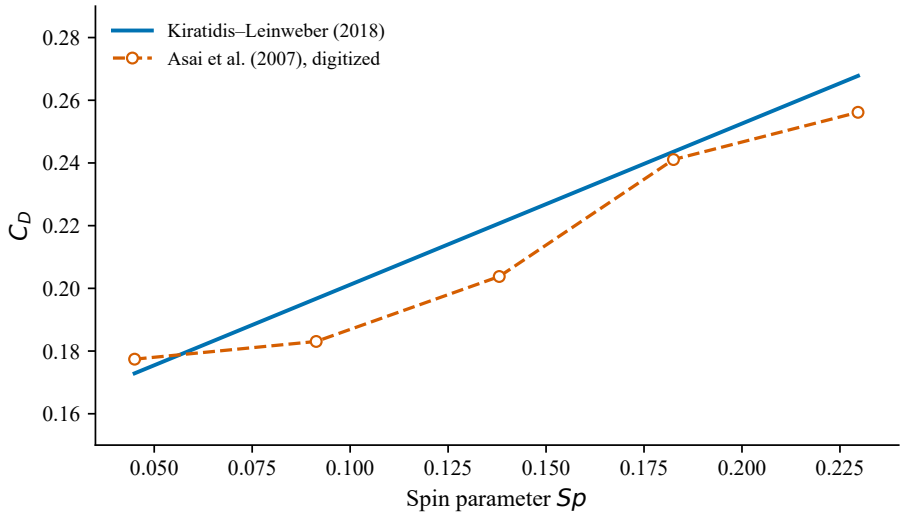


Figure 3: Teamgeist drag at $Re = 455172$: the complete empirical relation [8] and five digitized Asai measurements [7], Figure 8.

3.2.3 Magnus Lift

Rotation changes the flow on opposite sides of the ball and creates a transverse pressure force. Using the drag dynamic-pressure scale gives $F_L = qAC_L$ for transverse spin [11, 8]. For a three-dimensional spin axis, we adopt

$$\begin{aligned} \boldsymbol{F}_L &= \tfrac{1}{2}\rho A \alpha_L C_L(Re, Sp) v^2 \sin\theta \, \boldsymbol{e}_L, \\ \boldsymbol{e}_L &= \frac{\boldsymbol{\omega} \times \boldsymbol{u}}{\|\boldsymbol{\omega} \times \boldsymbol{u}\|}, \qquad \sin\theta = \frac{\|\boldsymbol{\omega} \times \boldsymbol{u}\|}{\omega v}. \end{aligned} \tag{6}$$

Lift is zero for zero spin, zero airspeed or parallel spin and airflow.

1. **Dynamic-pressure and area scale $\rho A v^2/2$.** Density, projected area and speed have the same meanings as in the drag force.

2. **Magnus direction e_L.** This unit vector is perpendicular to airflow. Topspin bends the path downward, backspin upward, and sidespin horizontally.

3. **Angular projection $\sin\theta$.** The transverse component is one for perpendicular spin and airflow and vanishes for parallel directions. This projection extends the transverse empirical law to an oblique spin axis.

4. **Lift coefficient $C_L(Re, Sp)$.** Rotation and boundary-layer state jointly determine lift.

   Rotation creates an asymmetric flow and a transverse Magnus force. Its magnitude depends on the boundary-layer state as well as spin, so a spin-only relation does not describe all flow conditions. Kiratidis and Leinweber [8] introduce the reference power law in their Eq. (8),

   $$C_L^{\text{fit}}(Sp) = 1.15 Sp^{0.83}, \tag{7}$$

   and scale it using drag as an indicator of flow state in their Eq. (9) [8]. We use that complete relation:

   $$C_L(Re, Sp) = C_L^{\text{fit}}(Sp) \frac{C_{D,\text{ceil}} - \min[C_D(Re, Sp), C_{D,\text{ceil}}]}{C_{D,\text{ceil}} - C_{D,\text{ref}}}, \tag{8}$$

   with the Teamgeist reference quantities

   $$C_{D,\text{ceil}} = f(0), \qquad C_{D,\text{ref}} = C_D(333793, 0.19). \tag{9}$$

   Here $C_{D,\text{ceil}} \simeq 0.4927$ is the zero-spin low-speed ceiling. The reference drag is evaluated with Eq. (5), including its spin term and using the same R and ν as the calculation. At the reference $(Re, Sp) = (333793, 0.19)$ the scaling factor is one and $C_L \simeq 0.290$. The ceiling, reference condition and use of drag to reduce lift follow Kiratidis and Leinweber [8]; Eq. (8) supplies the final lift coefficient in the force model.

5. **Lift correction $\alpha_L \geq 0$.** This dimensionless multiplier is one in forward prediction and is fitted to early positions when inverse prediction includes lift.

Comparison with published wind-tunnel data. Figure 4 uses the same Asai experiment and flow condition. The complete relation reproduces increasing lift with Sp and lies above all five measured values: MAE 0.04505, RMSE 0.05040 and maximum difference 0.07325. This larger discrepancy motivates the inverse lift correction α_L.

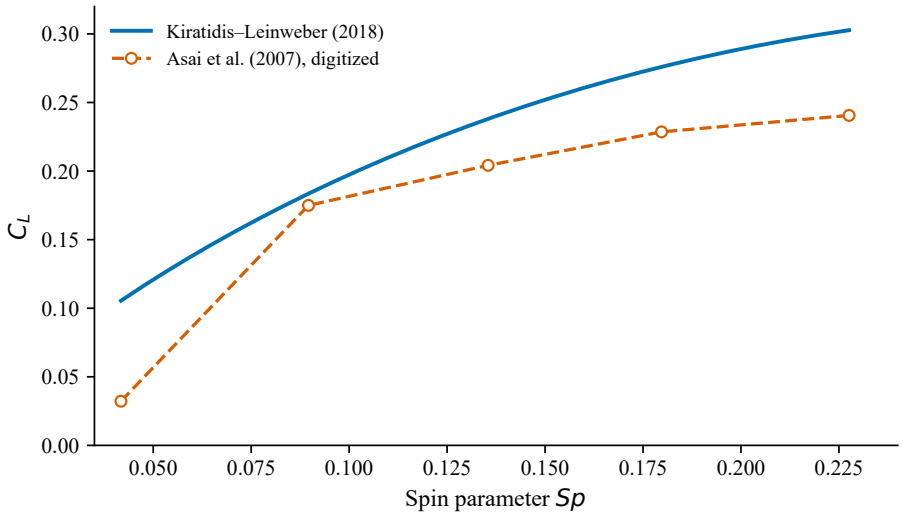


Figure 4: Teamgeist transverse lift at $Re = 455172$: the complete empirical relation [8] and five digitized Asai side-force measurements [7], Figure 9.

3.3 Dynamical Equations

Newton's second law and angular-momentum balance give

$$m\dot{\boldsymbol{v}} = \sum \boldsymbol{F}, \qquad \frac{d\boldsymbol{L}}{dt} = \boldsymbol{Q}, \qquad \boldsymbol{L} = I_b\boldsymbol{\omega}. \tag{10}$$

With zero applied torque and constant spherical inertia, the main model is

$$\begin{cases} \dot{\boldsymbol{r}} = \boldsymbol{v}, \\ m\dot{\boldsymbol{v}} = \boldsymbol{F}_g + \boldsymbol{F}_D + \boldsymbol{F}_L, \\ \dot{\boldsymbol{\omega}} = \boldsymbol{0}. \end{cases} \tag{11}$$

Its initial conditions are

$$\begin{cases} \boldsymbol{r}(0) = \boldsymbol{r}_0, \\ \boldsymbol{v}(0) = (v_{x0}, v_{y0}, v_{z0}), \\ \boldsymbol{\omega}(0) = (\omega_{x0}, \omega_{y0}, \omega_{z0}). \end{cases} \tag{12}$$

The observed initial position fixes the origin of the flight. The six velocity and spin components determine its initial motion. In still air, drag removes mechanical energy while Magnus lift does no work along the velocity; gravity converts kinetic and potential energy during flight.

4 Model Solving

We compare increasing levels of aerodynamic description to identify which effects control the trajectory. Table 3 defines the four models. Forward prediction uses measured or reconstructed launch conditions and unit force corrections. Inverse prediction uses two later positions to estimate the initial motion and force corrections, then propagates the fitted state to the goal.

Table 3: Four-model hierarchy. Gravity and constant spin are common to all models.

Model	Drag	Magnus lift	Inverse quantities
M1	none	none	$\boldsymbol{v}_0$ (3)
M2	constant C_D^*	none	$\boldsymbol{v}_0, \alpha_D$ (4)
M3	constant C_D^*	constant C_L^*	$\boldsymbol{v}_0$, spin axis, α_D, α_L (7)
M4	$C_D(Re, Sp)$	$C_L(Re, Sp)$	$\boldsymbol{v}_0, \boldsymbol{\omega}_0, \alpha_D, \alpha_L$ (8)

For M2 and M3, the constants are evaluated once at the published reference point:

$$C_D^* = C_D(333793, 0.19), \qquad C_L^* = C_L(333793, 0.19) = 1.15(0.19)^{0.83}. \tag{13}$$

For the competition air and ball parameters, $C_D^* = 0.2415$ and $C_L^* = 0.2898$. All models use the same physical conditions within each comparison. M1 and M2 have no spin-dependent trajectory; M3 uses the spin-axis direction, and M4 uses both its direction and magnitude.

4.1 Forward Prediction

4.1.1 Prediction procedure and experimental conditions

Given $\boldsymbol{r}_0$, $\boldsymbol{v}_0$ and $\boldsymbol{\omega}_0$, we set $\alpha_D = \alpha_L = 1$ and integrate Eq. (11) using an adaptive Runge–Kutta method. A root of $x - X_g$ locates the goal-plane crossing, and a descending root of $z - R$ locates ground contact. The calculation records speed and aerodynamic force magnitudes along the flight.

Goff and Carré [11] launched a 32-panel soccer ball using independently driven wheels and recorded its motion with two high-speed cameras. The reported ball mass is 0.424 kg, diameter 0.218 m, air density 1.2 kg/m^3 and kinematic viscosity 1.54×10^{-5} m^2/s. Rotation was measured by tracking marks on the ball. Ground markers provided landing ranges with a reported error of at most ± 0.05 m.

We use the nonspinning launch in their Figures 3, 4 and 11: approximate initial speed 18 m/s, elevation 22°, initial centre height about 0.85 m, and measured range 21.9 m. Four topspin launches from their Figure 14 provide additional spatial trajectory segments. Their published initial Re and Sp determine speed and spin; the first three digitized positions determine the initial spatial tangent. All five comparisons use red measurement-marker centres extracted from the vector PDF. The nonspinning early markers have 0.005 s spacing. Rotating segments are compared as $z(x)$ curves, matching their published spatial coordinates.

Table 4 specifies the forward inputs. All examples lie in the x–z plane, with $y_0 = v_{y0} = 0$ and $\omega_{x0} = \omega_{z0} = 0$; the listed v_{x0}, v_{z0} and ω_{y0} therefore complete the six initial velocity and spin components. Each first marker fixes the initial position.

Table 4: Initial conditions for the controlled forward examples, reconstructed as described above.

Launch	x_0 (m)	z_0 (m)	v_{x0} (m/s)	v_{z0} (m/s)	ω_{y0} (rad/s)
Zero spin	0.0001	0.8501	16.689	6.743	0.00
Topspin a	0.0001	0.8499	16.825	7.366	82.57
Topspin b	0.0001	0.8499	18.194	6.810	44.56
Topspin c	-0.0070	0.8526	17.383	8.188	125.16
Topspin d	-0.0067	0.8500	18.770	10.863	47.75

4.1.2 Calculated trajectories and physical interpretation

Table 5: Nonspinning controlled launch: predicted landing and signed range error. The reported range is 21.9 m.

Model	Range (m)	Prediction minus measurement (m)
M1	24.650	+2.750
M2	20.656	-1.244
M3	20.656	-1.244
M4	21.871	-0.029

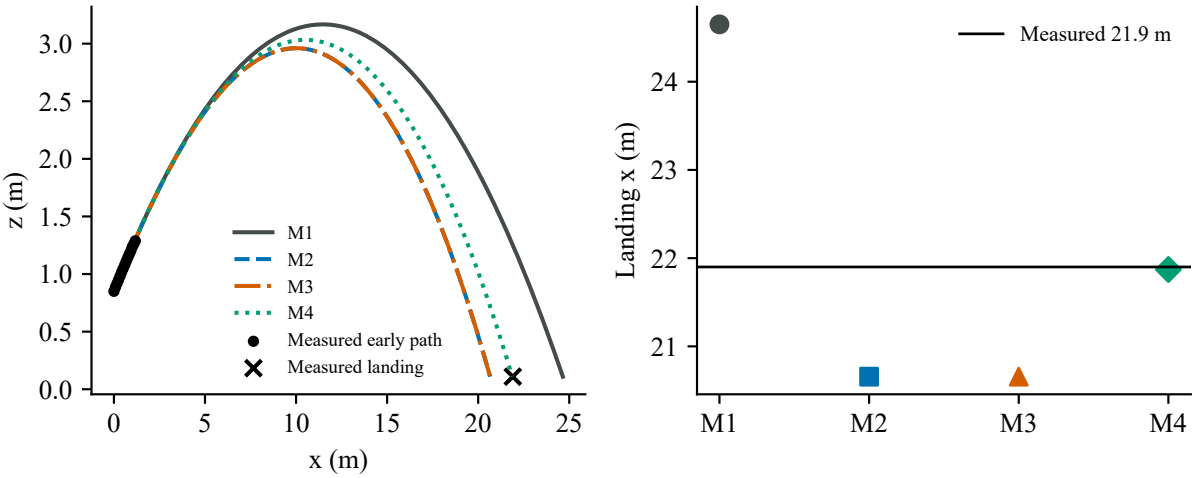


Figure 5: Forward prediction of Goff and Carré's nonspinning experiment [11]. Left: calculated flight paths, digitized early positions and measured landing. Right: individual model ranges against the measured 21.9 m. M2 and M3 coincide because the initial spin is zero.

Gravity alone produces the longest range. Air resistance removes horizontal momentum and shortens flight distance. M2 and M3 use the same constant drag and predict identical nonspinning trajectories. M4 allows drag to change with the slowing airflow; its predicted range differs from the measured value by -0.029 m under the stated approximate launch conditions.

Table 6: Height RMSE (m) at digitized horizontal positions for the four topspin segments.

Launch	M1	M2	M3	M4
a	0.0119	0.0118	0.0062	0.0072
b	0.0396	0.0397	0.0459	0.0466
c	0.0100	0.0100	0.0047	0.0099
d	0.0171	0.0170	0.0090	0.0082

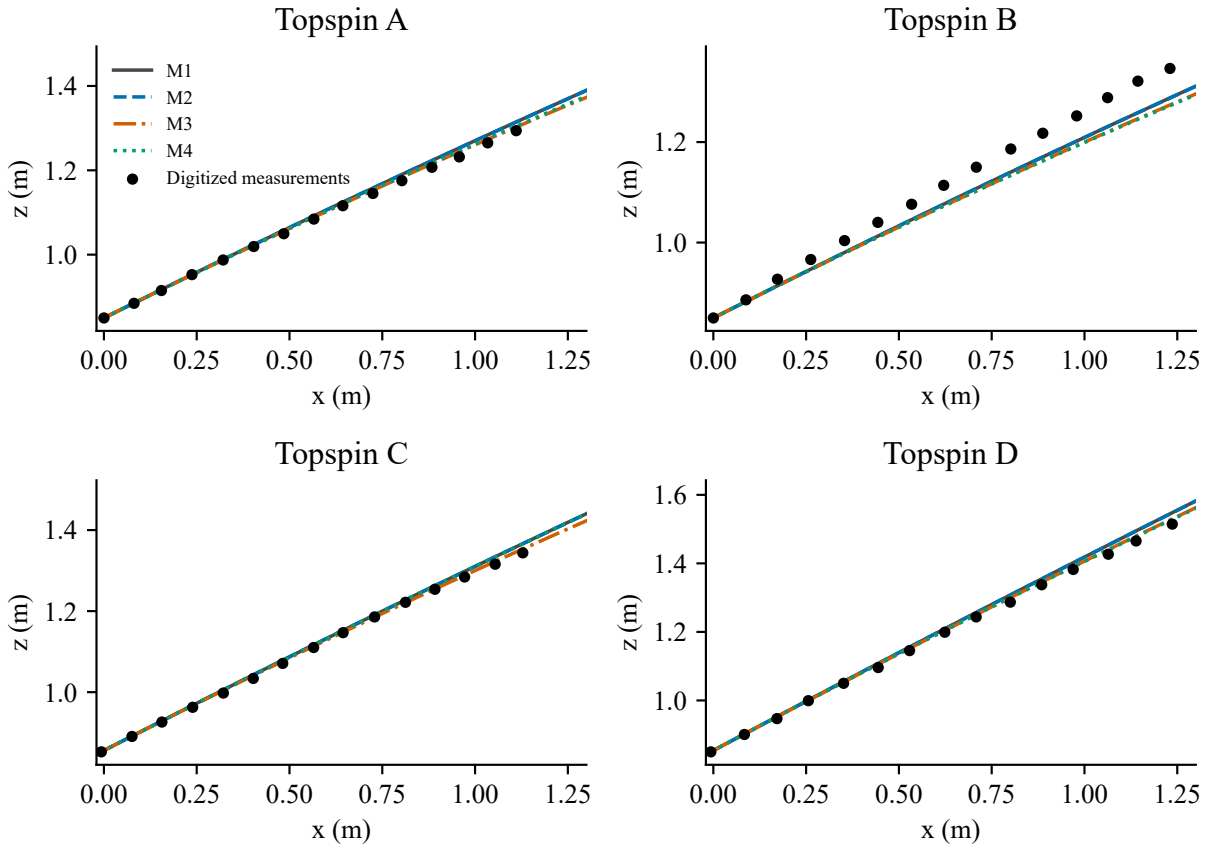


Figure 6: Four topspin launches digitized from Goff and Carré [11], Figure 14. Black points are measured marker centres; model curves use unit force corrections and the stated initial conditions. Each panel covers the published early spatial segment.

Topspin produces downward Magnus force and increases downward curvature. Table 6 gives the height RMSE at measured horizontal positions. The ranking varies across launches. Local direction estimates and the experimental ball's surface both influence the comparison with the Teamgeist reference. In topspin c, the high spin parameter raises the empirical drag above the ceiling in Eq. (8), so M4 assigns zero lift. The measured path retains downward spin-induced curvature, and M3 consequently gives a smaller local error in that case. These examples connect differences in the coefficient law to differences in the predicted path.

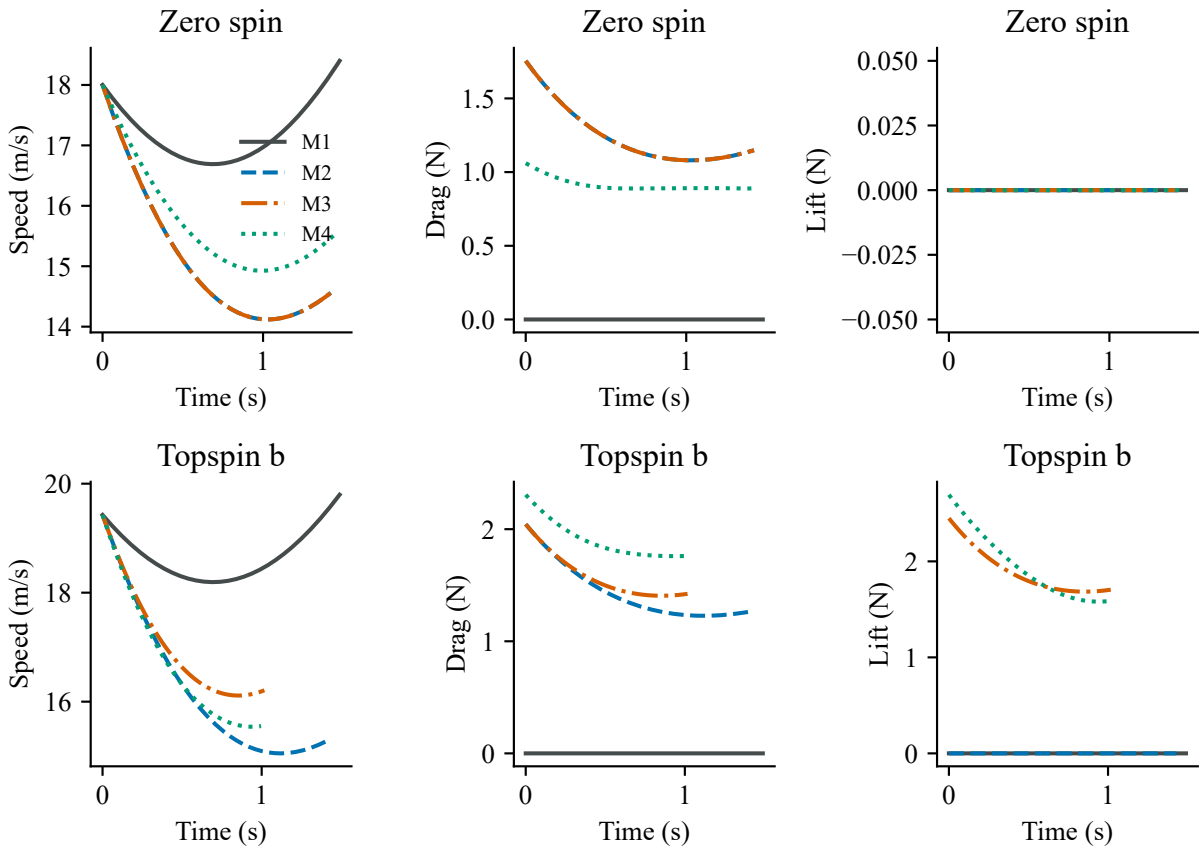


Figure 7: Speed, drag and lift during the nonspinning and topspin-b forward examples. Force magnitudes are calculated from the shared equations. Constant aerodynamic coefficients still produce changing forces because dynamic pressure varies with speed.

Figure 7 shows the mechanism behind these paths. Drag is strongest when the ball is fast. Downward lift in a topspin flight changes its vertical motion and flight duration. In M4, changing speed also changes Re and Sp, so the force evolution includes both dynamic-pressure and coefficient effects.

4.2 Inverse Prediction

4.2.1 Early-position fitting and representative solution

Let $\boldsymbol{p}_1$ and $\boldsymbol{p}_2$ be positions at t_1, t_2 after the known $\boldsymbol{r}_0$. The initial motion and applicable force corrections are chosen to minimize

$$J_r = \sum_{i=1}^{2} \|\boldsymbol{r}(t_i;\boldsymbol{\eta}) - \boldsymbol{p}_i\|^2, \qquad \boldsymbol{\eta}_{\text{M4}} = (\boldsymbol{v}_0, \boldsymbol{\omega}_0, \alpha_D, \alpha_L). \tag{14}$$

The next selection criterion favors the published force scale,

$$J_\alpha = (\alpha_D - 1)^2 + (\alpha_L - 1)^2, \tag{15}$$

using only the terms present in each model. Among equal-quality M4 states, the smallest spin magnitude selects a representative state. Deterministic multi-start optimization applies these criteria in order. The position-error allowance is 6×10^{-12} m^2, and the coefficient-objective allowance is 10^{-8}; both are numerical selection tolerances. Initial guesses use measured displacement velocities and the literature's experimental spin rates. M3 represents its spin axis by two angles; its unit vector serves only to carry direction in Eq. (6).

4.2.2 Comparison with recorded match flights

The Sportec Open DFL dataset associates optical tracking with match events [2, 3]. We analyze four flights from matches J03WOH and J03WQQ, and one recorded penalty. Events are selected for an identifiable shot and a tracked crossing of the actual goal-line plane. The four flights comprise two headers and two foot shots. Each inverse calculation uses the first three available positions at 0, 0.04 and 0.08 s. Later tracking coordinates determine the observed crossing of the goal plane. The penalty segment begins with a slower first displacement before the faster recorded flight; its early coordinates are used as recorded.

Table 7: All five recorded flights and all four inverse models. Error is the Euclidean difference in goal-plane (y, z).

Event	Model	Termination	Early residual (m)	Plane error (m)
WOH 0487 (head)	M1	plane	0.036	0.717
WOH 0487 (head)	M2	plane	0.00501	2.348
WOH 0487 (head)	M3	plane	2.28×10^{-12}	1.909
WOH 0487 (head)	M4	ground	3.53×10^{-10}	—
WOH 0606 (foot)	M1	plane	0.0263	4.988
WOH 0606 (foot)	M2	ground	0.0303	—
WOH 0606 (foot)	M3	ground	1.32×10^{-12}	—
WOH 0606 (foot)	M4	ground	1.83×10^{-10}	—
WOH 0968 (foot)	M1	ground	0.124	—
WOH 0968 (foot)	M2	ground	0.124	—
WOH 0968 (foot)	M3	early termination	0.127	—
WOH 0968 (foot)	M4	plane	0.126	10.893
WQQ 1227 (head)	M1	plane	0.18	2.785
WQQ 1227 (head)	M2	plane	0.172	3.325
WQQ 1227 (head)	M3	fit failure	0.0226	—
WQQ 1227 (head)	M4	fit failure	0.0952	—
Penalty	M1	plane	0.204	0.853
Penalty	M2	plane	0.204	0.853
Penalty	M3	fit failure	0.208	—
Penalty	M4	fit failure	0.208	—

Table 8: Reached-case errors and termination counts. Individual errors are retained in the preceding table.

Model	Reached	Ground	Other	Fit failures	Mean error (m)
M1	4/5	1	0	0	2.336
M2	3/5	2	0	0	2.175
M3	1/5	1	1	2	1.909
M4	1/5	2	0	2	10.893

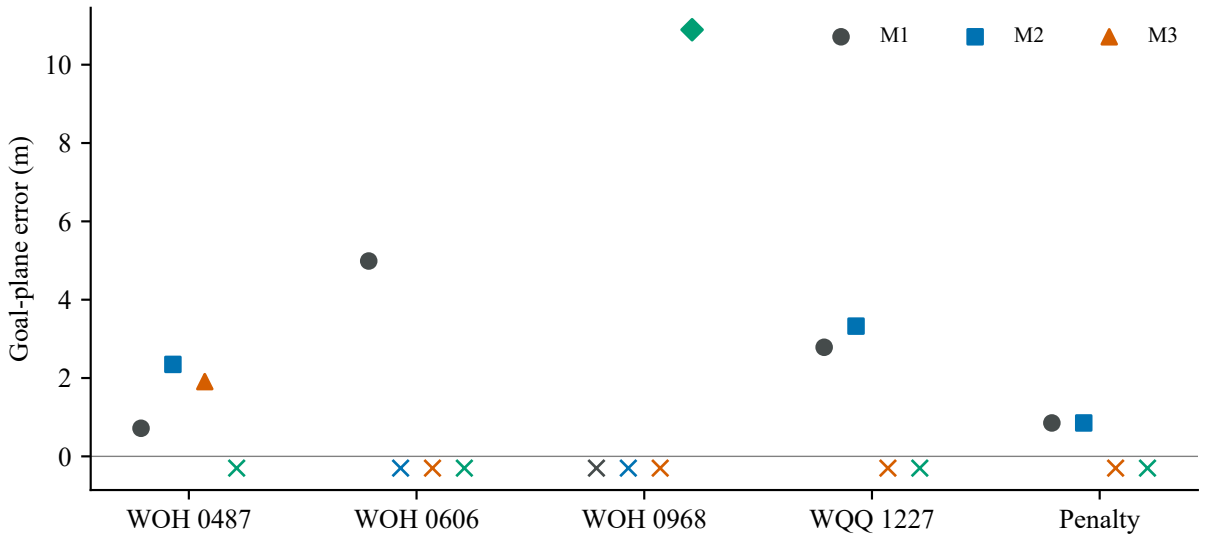


Figure 8: Individual goal-plane prediction errors for the five recorded flights. Each marker is one event and model. Crosses below zero denote trajectories that provide no goal-plane prediction; their termination status appears in Table 7.

M1 reaches the goal plane in 4 of five cases; M2 reaches the goal plane in 3 of five cases; M3 reaches the goal plane in 1 of five cases; M4 reaches the goal plane in 1 of five cases. M1 gives the largest number of goal-plane predictions in these five recorded flights. The table reports each reached-case mean together with its contributing count. Large force corrections and iteration-limit cases reveal a strong dependence on the early recorded motion. Fit failures denote numerical iteration limits; their candidate trajectories are retained in the calculation records.

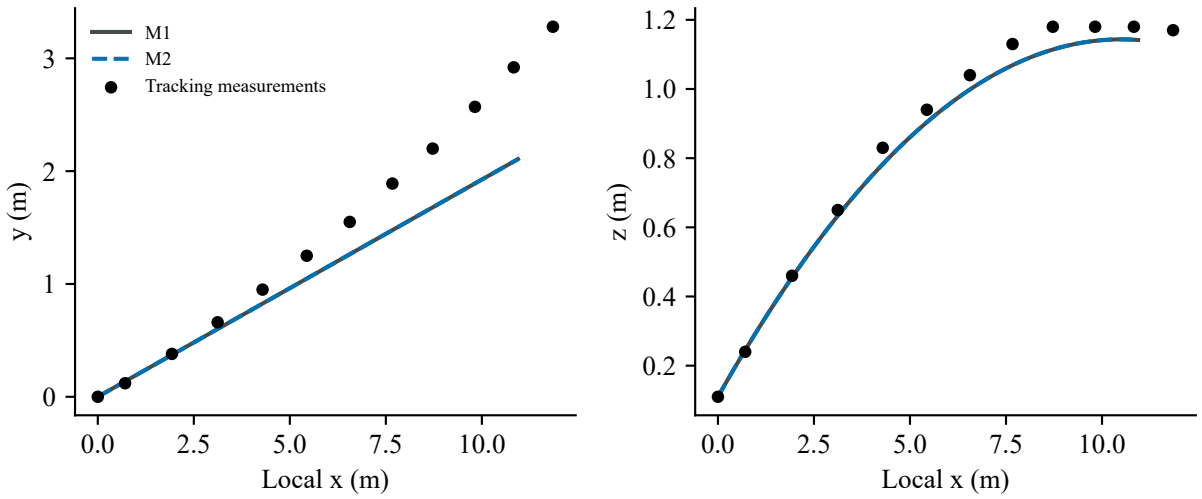


Figure 9: Converged inverse predictions for the recorded penalty, with all tracked positions shown. The first three positions enter the fit; remaining positions display the observed continuation. All model statuses are listed in Table 7.

The comparison links early acceleration to later flight. A large fitted drag correction produces rapid momentum loss, while lift corrections alter the direction of curvature. Changes in early recorded displacement therefore have a strong effect on the predicted endpoint. The event-level results show how each model responds to the actual recorded segment.

4.2.3 Competition Problem B

The supplied observations are

t (s)	x (m)	y (m)	z (m)
0	0	0	0.11
0.05	1.50	0.45	0.44
0.10	2.96	0.86	0.74

(16)

Applying the same inverse procedure gives Table 9. The preferred M4 initial conditions are

$$\begin{aligned} \boldsymbol{v}_0 &= (30.406, 9.426, 6.906)\ \mathrm{m/s}, \\ \boldsymbol{\omega}_0 &= (-49.090, -31.483, -112.266)\ \mathrm{rad/s}, \\ (\alpha_D, \alpha_L) &= (1.0001, 1.0000). \end{aligned} \tag{17}$$

Table 9: Inverse predictions for the competition observations. Residual is the maximum absolute early-coordinate difference.

Model	α_D	α_L	Residual (m)	y_g (m)	z_g (m)	t_g (s)
M1	—	—	0.016	3.217	1.957	0.3706
M2	1.5683	—	0.01	3.217	1.879	0.4058
M3	1.5666	1.0000	1.5×10^{-12}	2.661	1.938	0.4008
M4	1.0001	1.0000	6.4×10^{-13}	2.683	1.959	0.4026

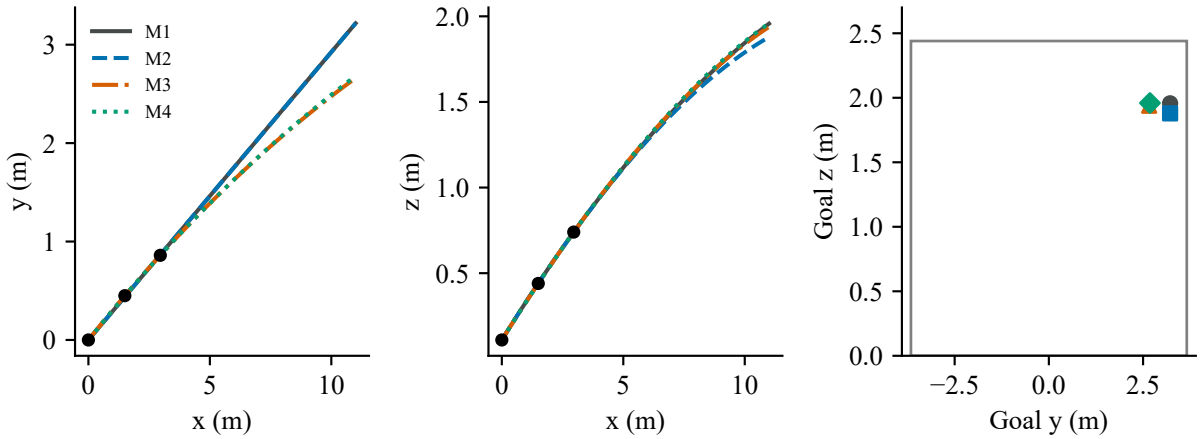


Figure 10: Four inverse predictions for Problem B: horizontal path, vertical path and ball-centre positions at the goal plane. Black points are the supplied early observations.

M1 and M2 predict nearly the same horizontal direction because gravity and drag preserve the horizontal velocity ratio. Adding Magnus lift in M3 and M4 bends the ball laterally. The M4 goal-plane centre is $(2.683, 1.959)$ m at 0.4026 s and fits inside the goal with the ball-radius margin. Its early maximum coordinate residual is 6.4×10^{-13} m. The inferred initial spin is 126.51 rad/s; it supplies the aerodynamic curvature required by this representative fit.

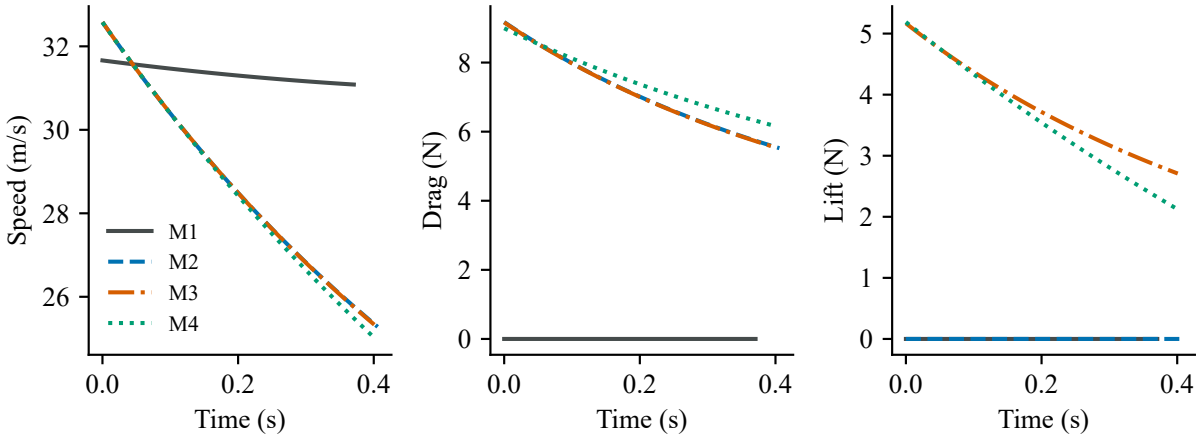


Figure 11: Speed and aerodynamic force magnitudes for the four fitted Problem B models. All share the same observations, while their force assumptions produce different continuation paths.

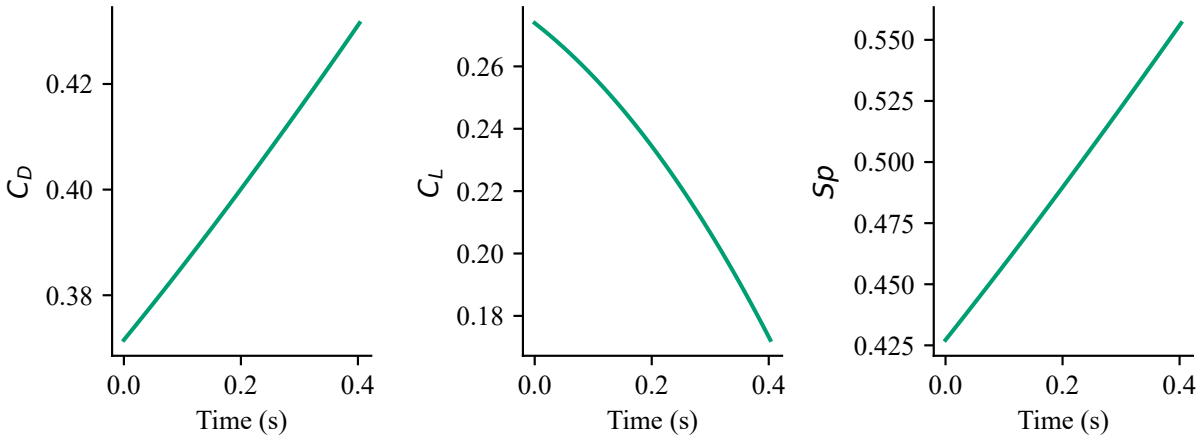


Figure 12: The complete M4 aerodynamic coefficients and spin parameter during the Problem B flight. Constant spin combined with decreasing speed raises Sp; the coupled empirical relations determine the resulting drag and lift coefficients.

The coefficient evolution in Figure 12 shows why constant spin still produces changing aerodynamics. The surface-to-air speed ratio increases as the ball slows, while the drag-dependent scaling also changes the lift coefficient.

5 Sensitivity Analysis

To be completed.

6 Discussion of Other Factors

The main model isolates gravity, drag and Magnus lift. We now examine the physical effects omitted from its geometry, spin evolution and ambient airflow.

6.1 Asymmetric Surface Geometry

Paudel and Yan [5] performed full-flight flow simulations using realistic football geometry and a smooth-sphere comparison. For launch speeds of 10, 15 and 20 m/s and spin rates of 31.5, 63 and 94.5 rad/s, their results show that seams and surface structure alter separation, wake dynamics and Magnus force. The smooth sphere produces greater lateral bending in their nine comparisons, and vertical flight-time differences become more visible at higher speed and rotation.

The spherical geometry in the present report describes the centre-of-mass motion and projected area. Its force coefficients come from football measurements, so the average surface influence enters through C_D and C_L. Explicit panel orientation and asymmetric wake fluctuations provide a further level of description when lateral-force details are important. These published results support retaining football-specific coefficients in a spherical trajectory model.

6.2 Rotation and Aerodynamic Damping

Rotation changes Magnus force, so a declining spin can change the continuation of a curved shot. According to Grad et al. [9], Supplementary Section 8.4, and the MuJoCo ellipsoid model [10], the spherical inertial and viscous damping torques are

$$\boldsymbol{Q} = -k_2\omega\boldsymbol{\omega} - k_1\boldsymbol{\omega}, \quad k_2 = \frac{8\pi}{15}\rho C_{\text{angular}} R^5, \quad k_1 = 8\pi\mu R^3. \tag{18}$$

The quadratic term describes high-Reynolds-number angular drag; the linear term is viscous resistance. We use MuJoCo's documented reference value $C_{\text{angular}} = 1.5$ [10]. Defining $q = \rho v^2/2$ and $C_{Q,v} = \|\boldsymbol{Q}\|/(qAR)$ gives

$$C_{Q,v} = \frac{16}{15} C_{\text{angular}} Sp^2 + \frac{32Sp}{Re}, \qquad C_{Q,\omega} = \frac{C_{Q,v}}{Sp^2} = \frac{16}{15} C_{\text{angular}} + \frac{32}{Re\, Sp}. \tag{19}$$

$C_{Q,\omega}$ uses rotating surface speed $R\omega$ in its reference pressure. The factor $1/Sp^2$ converts between the two normalizations. The viscous-to-inertial torque ratio is

$$\frac{Q_{\text{viscous}}}{Q_{\text{inertial}}} = \frac{30}{C_{\text{angular}} Re\, Sp} = \frac{15\nu}{C_{\text{angular}} R^2 \omega}. \tag{20}$$

It is 9.8×10^{-5} at the fitted competition launch, so the quadratic term dominates. Following the thin-shell inertia used by Maponi and Piombin [4], $I_b = 2mR^2/3$. Adding the high-Reynolds-number torque gives

$$\begin{cases} \dot{\boldsymbol{r}} = \boldsymbol{v}, \\ m\dot{\boldsymbol{v}} = \boldsymbol{F}_g + \boldsymbol{F}_D + \boldsymbol{F}_L, \\ I_b\dot{\boldsymbol{\omega}} = -k_2\omega\boldsymbol{\omega}. \end{cases} \tag{21}$$

The torque opposes spin and removes rotational energy. With this isolated quadratic law, $\omega(t) = \omega_0/[1 + (k_2/I_b)\omega_0 t]$, providing an integration check.

We repeat the M4 Problem B flight with exactly its fitted initial state and force corrections, first with constant spin and then with Eq. (21). The state is held fixed so the comparison measures the damping effect directly.

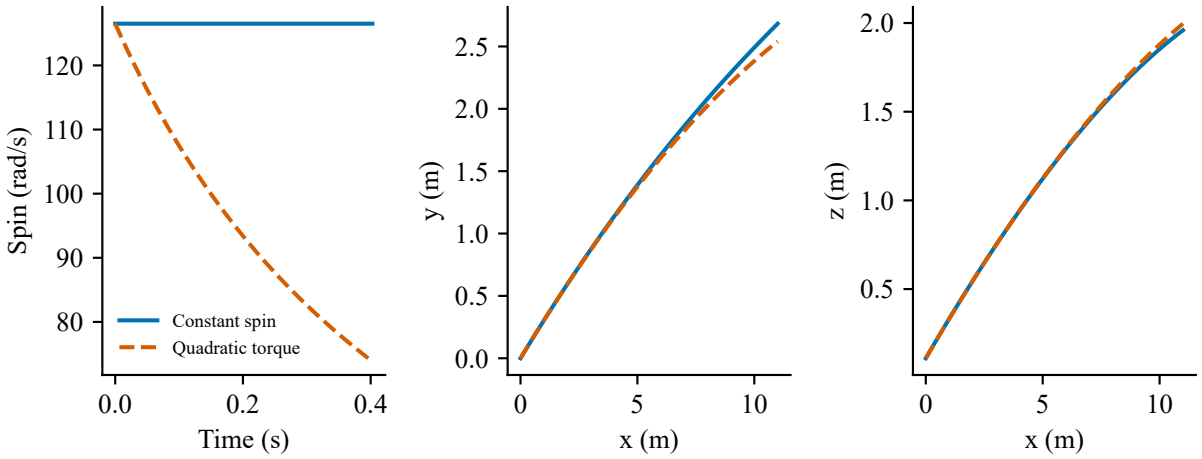


Figure 13: Same-initial-state comparison of constant spin and high-Reynolds-number rotational damping for Problem B. The documented angular-drag reference is 1.5, and the ball uses thin-shell inertia.

The damped goal-plane centre is $(2.541, 1.996)$ m, compared with $(2.683, 1.959)$ m at constant spin. The displacement is 0.147 m, or 1.34% of the 11 m flight distance. Spin decreases by 41.24%, and the goal-crossing time changes by -1.60%. Adding the much weaker viscous term changes the damped crossing by 6.5e-06 m. Thus this reference torque has a visible effect on spin and a smaller, measurable effect on the short-flight endpoint.

6.3 Wind Speed and Force Estimates

Wind changes the airflow through $\boldsymbol{u} = \boldsymbol{v} - \boldsymbol{v}_{\text{wind}}$ [4]. At fixed coefficients, a wind speed w along the flight direction changes the dynamic-pressure scale by

$$\frac{q_{\text{head}}}{q_0} = (1 + w/v)^2, \qquad \frac{q_{\text{tail}}}{q_0} = (1 - w/v)^2, \qquad \frac{q_{\text{cross}}}{q_0} = 1 + (w/v)^2. \tag{22}$$

Crosswind also rotates the airflow by $\arctan(w/v)$, changing the force direction. Table 10 evaluates these rough estimates at the fitted initial speed. The listed wind speeds are illustrative conditions for comparison.

Table 10: Illustrative wind estimates at the fitted launch speed. Percentages describe dynamic pressure at fixed aerodynamic coefficients.

w (m/s)	Headwind (%)	Tailwind (%)	Crosswind (%)	Airflow angle (°)
1	+6.23	-6.05	+0.09	1.76
3	+19.27	-17.57	+0.85	5.26
5	+33.06	-28.34	+2.36	8.73

A 5% limit on the headwind dynamic-pressure change corresponds to $w/v \leq \sqrt{1.05} - 1$, or $w \leq 0.80$ m/s at this launch. The still-air approximation is most suitable for indoor or weak-wind flight; stronger wind should enter through the relative-velocity calculation.

7 Conclusions

The three-force model connects soccer-ball flight to three distinct mechanisms: gravity produces downward curvature, drag removes momentum, and Magnus lift changes direction. The complete Teamgeist coefficient relations represent flow-state and spin changes, while the matched 2007 wind-tunnel comparison gives a closer reproduction for drag than for lift.

The controlled nonspinning example shows that including drag is essential for range prediction. Its M4 range is 21.871 m against the published 21.9 m. The rotating trajectory segments display the downward curvature of topspin and variation in the accuracy of the common coefficient law. Recorded match comparisons show how early displacement and fitted force corrections affect subsequent prediction.

For Problem B, M4 gives $(y, z) = (2.683, 1.959)$ m at the goal plane. Magnus lift accounts for the marked horizontal difference from the gravity-only and drag-only predictions. The same-state rotational-damping comparison shifts the crossing by 0.147 m under the documented torque setting. Wind estimates give the weak-wind condition for the still-air approximation. Sensitivity-analysis findings will be added after Section 5 is completed.

8 Model Evaluation

8.1 Physical Interpretation and Computational Use

The hierarchy makes the contribution of each force visible. The same dynamics handles forward calculation, early-position inversion and endpoint prediction. Force corrections provide a direct way to accommodate measured trajectory differences, and all experimental and numerical inputs are retained with their sources.

8.2 Predictive Performance and Improvements

The wind-tunnel comparisons, controlled launches and recorded match flights examine complementary aspects of the model: coefficient trends, curvature and range, and prediction from early observations. Their event-level errors identify where a common football coefficient law and an

early trajectory segment give a useful continuation. Ball-specific aerodynamic measurements, directly measured spin and synchronized higher-rate positions would improve this connection. Explicit surface orientation, rotational damping and wind can be included as the application requires.

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Reproduction Code

The accompanying project contains the frozen inputs, vector-marker extraction, four-model calculation, result tables and figure scripts. The following excerpts show the shared state equation and force evaluation. The inverse solver uses the same state derivative with the three selection criteria described in Section 4.2. Calculation and figure generation are separate commands, allowing plots to be refreshed from saved results.

```
--
def state_derivative(t, state, config):
    """The single nine-state RHS used for prediction and inverse fitting."""
    f = forces(state[3:6], state[6:9], config)
    return np.r_[state[3:6], sum(f.values())/config.mass,
                 rotational_torque(state[6:9], config)/config.moment_of_inertia]
```

```
--
def forces(velocity, spin, config):
    """Return force components on ball, in newtons."""
    v = np.asarray(velocity, dtype=float) - np.asarray(config.wind, dtype=float)
    omega = np.asarray(spin, dtype=float)
    speed = np.linalg.norm(v)
    weight = np.array([0., 0., -config.mass * config.gravity if config.use_gravity else
        0.])
    drag = np.zeros(3)
    magnus = np.zeros(3)
    if speed > 1e-12:
        cd, cl, _, _, _ = coefficients(velocity, spin, config)
        q_area = 0.5 * config.air_density * np.pi * config.radius**2 * speed**2
        if config.use_drag:
            drag = -q_area * config.drag_scale * cd * v / speed
        cross = np.cross(omega, v)
        norm = np.linalg.norm(cross)
        if config.use_magnus and norm > 1e-12:
            # The literature fit is for transverse spin. Project its effect
            # onto the plane perpendicular to airflow for an oblique spin axis.
            sin_angle = norm / (np.linalg.norm(omega) * speed)
            magnus = q_area * config.lift_scale * cl * sin_angle * cross / norm
    return {"gravity": weight, "drag": drag, "magnus": magnus}
```